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Mechanisms and Models of Climate Variability.

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Points of View

(Or the reason we are funded to do science may differ from our personal motivation in doing it.)

- Predicting the behaviour of the climate system is terribly important to society.
  - Of order \$100 billion a year (estimates differ widely) to substantially cut CO_2 emissions (c.f., \$10 trillion economy). The cost of the alternative (living with global warming) may be much higher.
- It is also fun, interesting and intellectually challenging to understand the climate system.
  - It is almost uniquely a pure *and* an applied science at the same time. Fields from applied mathematics and turbulence to chemistry and molecular biology thrown into a big vat and stirred.

Justifying Research

Perhaps it is just cheap...

> *Basic research may seem very expensive. I am a well-paid scientist. My hourly wage is equal to that of a plumber, but sometimes my research remains barren of results for weeks, months or years and my conscience begins to bother me for wasting the taxpayer's money.... Basic research, to which we owe everything, is relatively very cheap when compared with other outlays of modern society. If one were to add up all the money ever spent by man on basic research, one would find it to be just about equal to the money spent by the Pentagon last year.*

Albert Szent-Gyorgyi (1893-1984), biochemist

Areas of science and humanities are justified in different ways....

Mathematics

- *Mathematics is increasingly vital to science, technology, and society itself. "The mathematical underpinnings of the computer revolution....the sophisticated mathematical design of the fuel-efficient Boeing 767 and European Airbus airfoils further exemplify the increased impact of mathematics.*"

  The David Report (1984), emphasizing the importance of mathematics to science and to society.

- *"The mathematician's patterns must be* beautiful*... . I have never done anything useful."*

  G. H. Hardy, (famous) mathematician, *A Mathematician's Apology*, 1940 (stated proudly, humbly, perhaps flippantly)

  "The world sickens at such cloistral clowning."

  Frederick Soddy, (famous) chemist, in reaction to Hardy.

A nuanced view from Poincaré

"Mathematics is not important because it helps us build machines; machines are important because they give us more time for mathematics."

Henri Poincaré, *Science and Method*, 1908.

Cosmologists

Have interesting views as to what science is about....

> *I don't see the logic of rejecting data just because they seem incredible.*
>
> Fred Hoyle, British Cosmologist.

and are sometimes perceived as arrogant....

> *Cosmologists are often wrong, but never in doubt.*
>
> Lev Landau, Russian Physicist.

The Practice of Climate Science

What is it and how is it studied?

- A branch of the earth sciences.
- A very complex dynamical system, with aspects of physics, chemistry, biology and a flavour unto itself.
- Study using:
  1. General Circulation Models on large computers. Incorporate all of the ‘microscopic’ laws of nature we know. The computer model integrates this, in both time and in concept.
  2. Theoretical (mathematical) models. Try to predict macroscopic behaviour ab initio. Supposedly leads to ‘understanding.’
  3. Multiple levels of model in between.
- Test using observations (present and past), from observations daily basis (weather forecasts!) to paleorecords.

Climate Questions, Scientific (and Philosophic?) Problems

Do we *need* to 'understand' in order to predict or to simulate with a GCM?
Yes and no — problem dependent.

Points of View

1. If we can simulate it (or predict it) we *do* understand it. A GCM is just a calculating tool that synthesizes our understanding of the underlying physics and chemistry.
   - Any additional kind of so-called understanding is a mental indulgence, a conceit.
2. Real understanding requires a mental picture — a concept — of a phenomenon.
   - If you can't explain it, you don't understand it.
3. Understanding a complex phenomenon requires a description at many levels of complexity.

Climate Questions - Examples

- Atmospheric heat transport from equator to pole.
  - This is calculated by a GCM far better than we conceptually understand it. The latter flounders for the lack (inter alia) of a theory of turbulence.
- But GCMs don't simulate ice ages well, or moist convection and the 'Intertropical Convergence Zone', or Gulf Stream separation, and estimates of global warming widely differ ($2^\circ - 6^\circ$ for CO_2 doubling.)
  - Sometimes an issue of model resolution (need faster computers), sometimes some physics or dynamics is presumably wrong or misrepresented in the model.
- So to take the next step, whatever than may be, we surely need a conceptual understanding too.
  - It's a dangerous game to simulate with no other level of understanding.

Specific Questions:

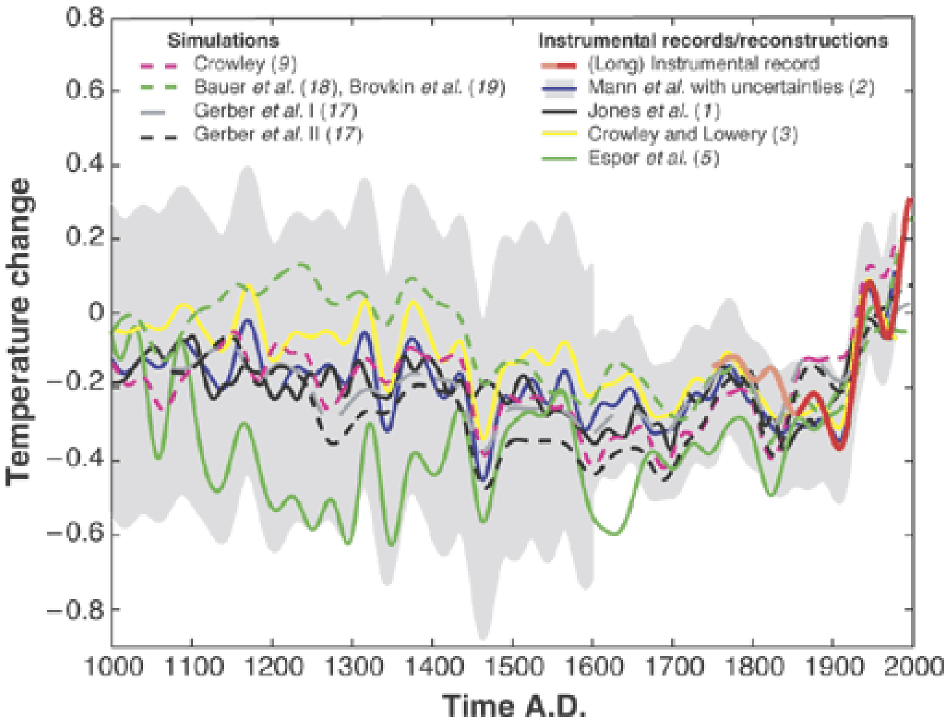
- What is the nature of climate variability on timescales from weeks to millenia?
  - What processes are involved? Is climate variability internal to the system, or externally forced?
- Is Global Warming going to change all that?
  - Is natural variability larger or smaller than anthropogenic climate change? On what timescales?

Past Climate Variations

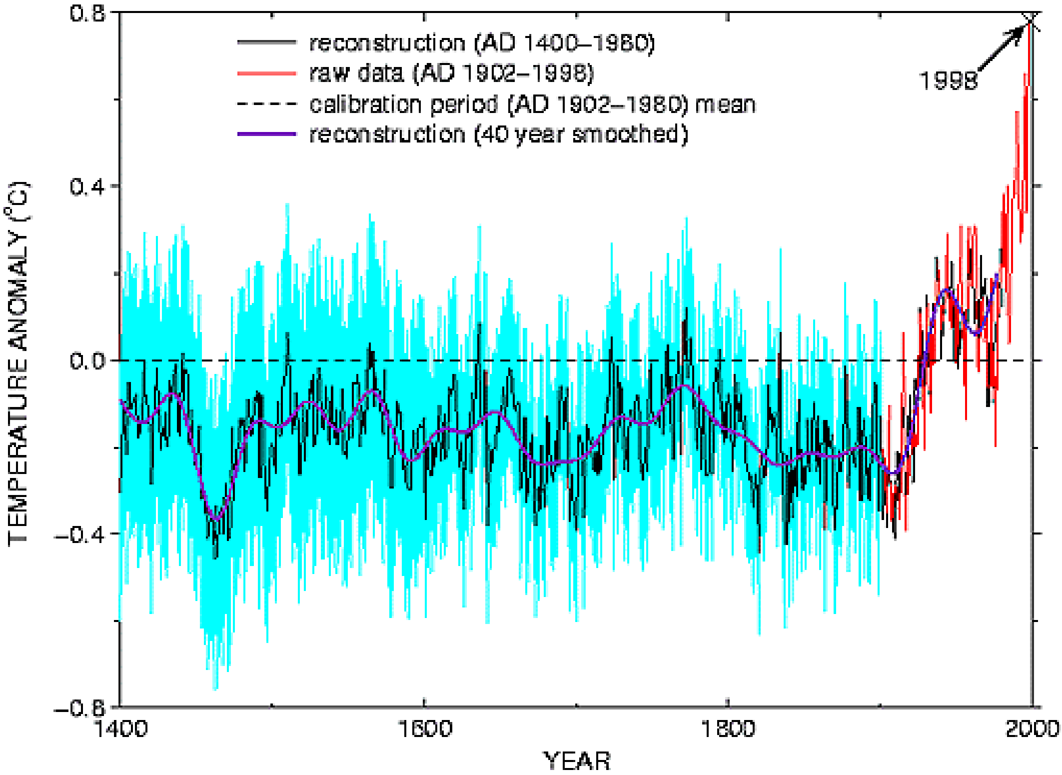
Previous climates were variously colder, warmer, and more variable than todays. The climate of the last few thousand years has been quite *stable* by historical standards.

Climate Variations

Temperature over the past millennium, reconstructed from multiple records and proxies (M. Mann).

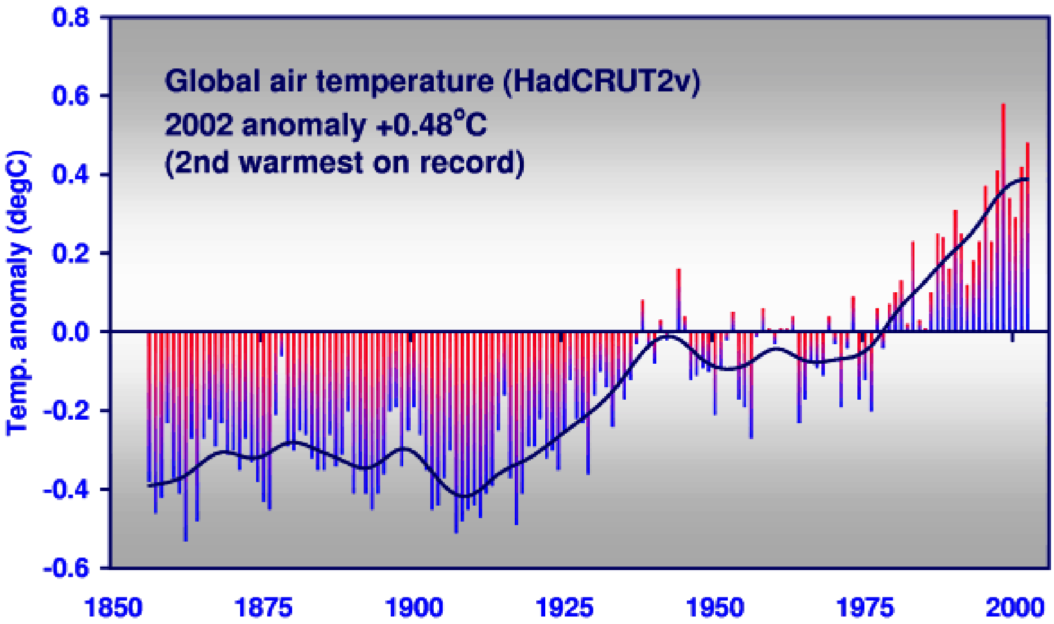


Temperature over the past 600 years.



Climate Variations

Temperature over the past 150 years
(Univ. East Anglia and UK Met Office)



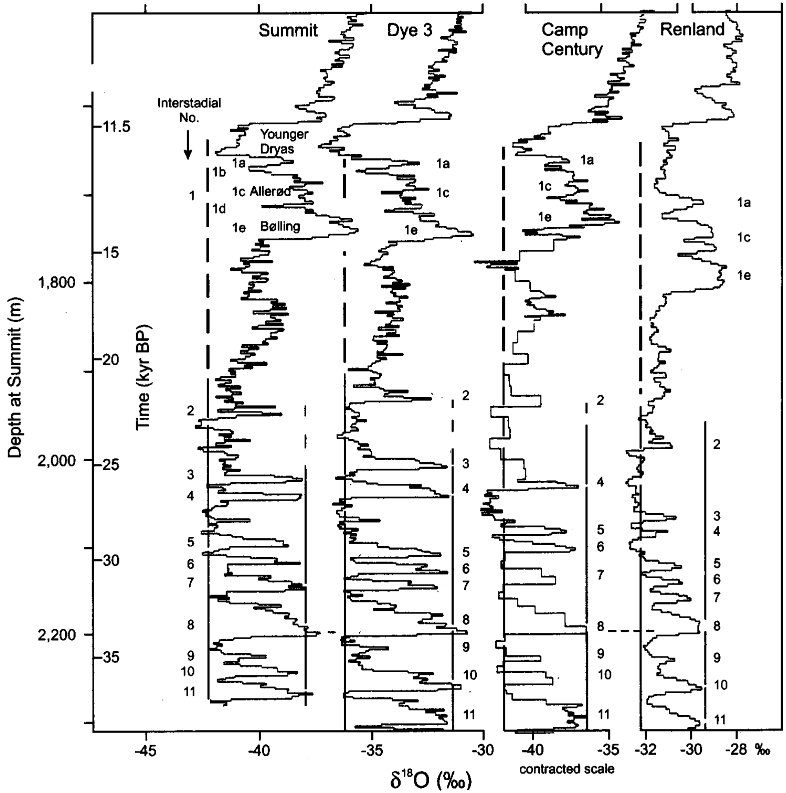
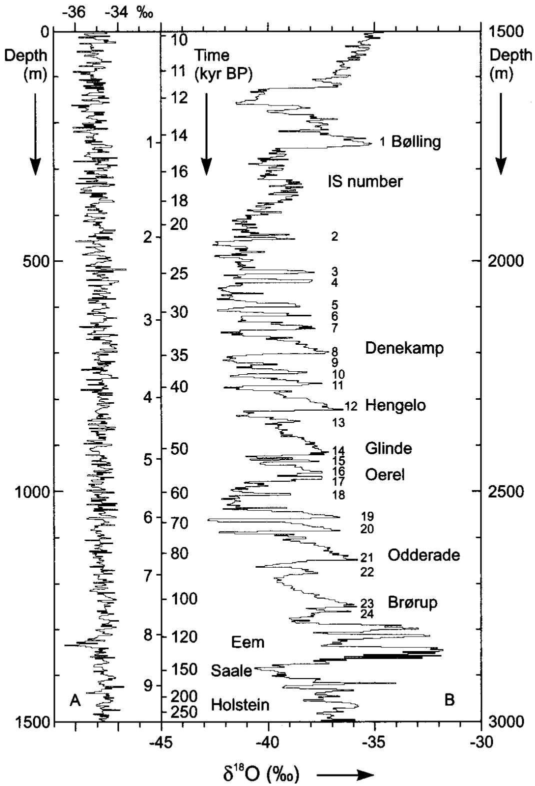


FIGURE 5.23 The $\delta^{18}O$ records from four Greenland sites (see Fig. 5.1) showing parallel, large-amplitude changes over very short periods of time. Changes are commonly "saw-tooth" in pattern (see Stage 8, for example) with an abrupt shift to higher $\delta^{18}O$ levels, followed by a slower return to lower values (Johnsen *et al.*, 1992).



δ^{18}O profile for GRIP ice core in Greenland. Lower (more negative) values of δ^{18}o indicate colder temperatures. The left panel is for the holocene period, and covers the last 10,000 years. This is evidently much less variable than the preceeding 200,000 years.

Timescales of Climate Change

- On very long timescales natural variability, in particular that associated glacial cycles, is probably larger than anthropogenic climate change.
  - Although if atmospheric CO_2 increases by eightfold, all bets are off.
- On interannual timescales, natural climate variability is larger than anthropogenic climate change.
  - El Niño is a significant cause of interannual variability, possibly variations in the thermohaline circulation the 'North Atlantic Oscillation', and volcanic activity.
- On decadal-century timescale, global warming is likely to be the dominant factor in climate change.
  - But any given decade need not be warmer than the previous. We are now emerging from the 'climate noise' of the last millennium.

Why is the Glacial Climate more Variable?

Clues to what might be going on

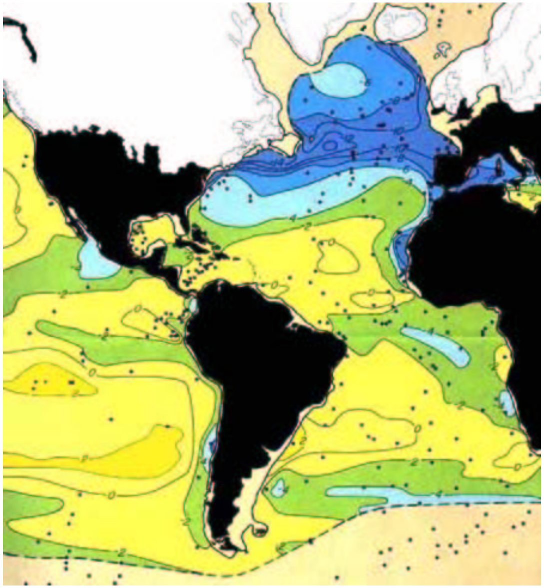
- The thermohaline circulation is most sensitive to the temperature gradient where the THC surfaces — primarily in the subpolar gyre and other high latitudes.

  In particular, the temperature gradient across the subpolar gyre plays an important role in the strength of the thermohaline circulation

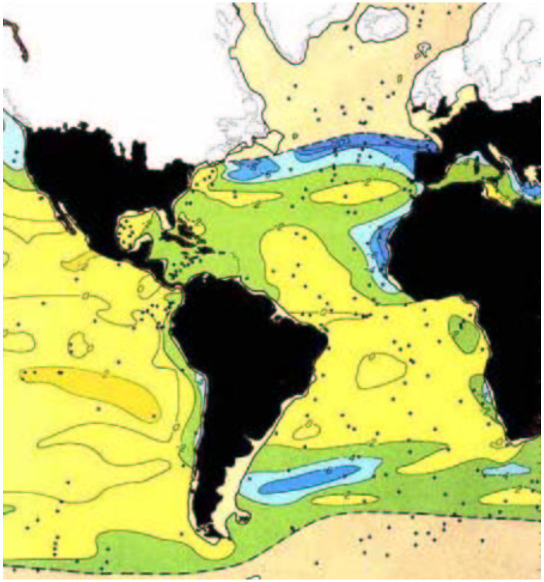
- If ice covers the subpolar gyre, the temperature gradient seen by the ocean is *weaker* than otherwise.
- A weak thermohaline circulation is more susceptible to irregular oscillations ('flushes') than a strong thermohaline circulation. (Various modelling and theoretical studies.)
- Some evidence (e.g. Oppo and Lehman) that warm (glacial time) SSTs are coincident with increased NADW production during D-O events.

SST: Difference between Last Glacial Maximum (CLIMAP), and Present Day

August

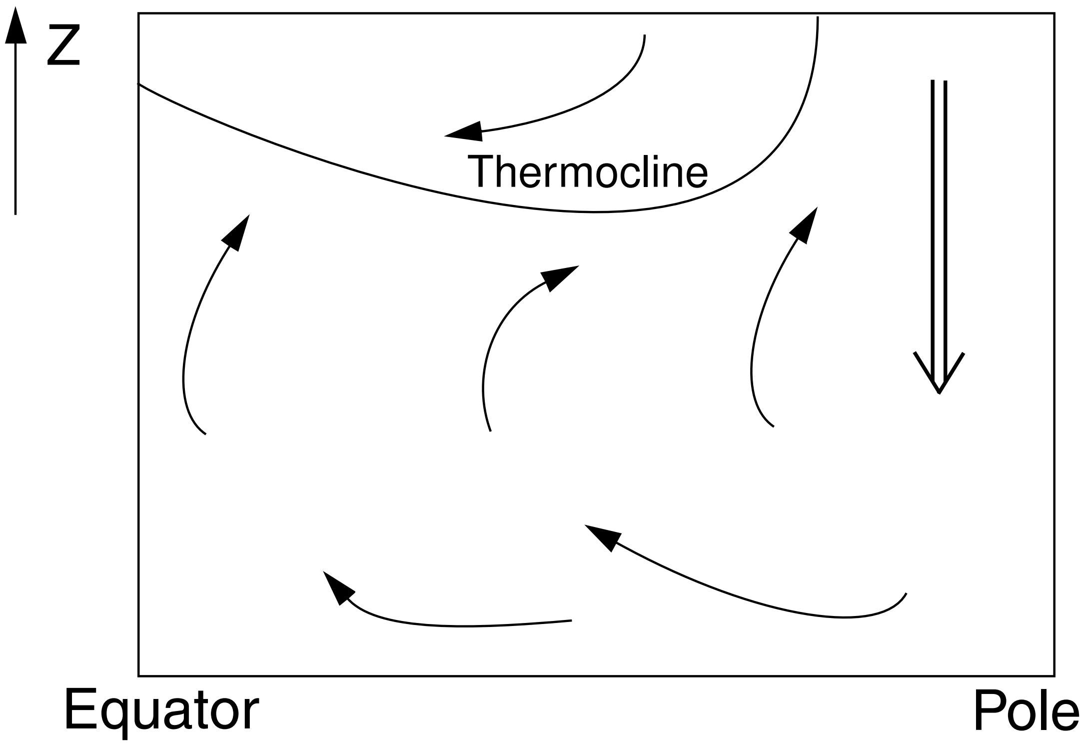


February



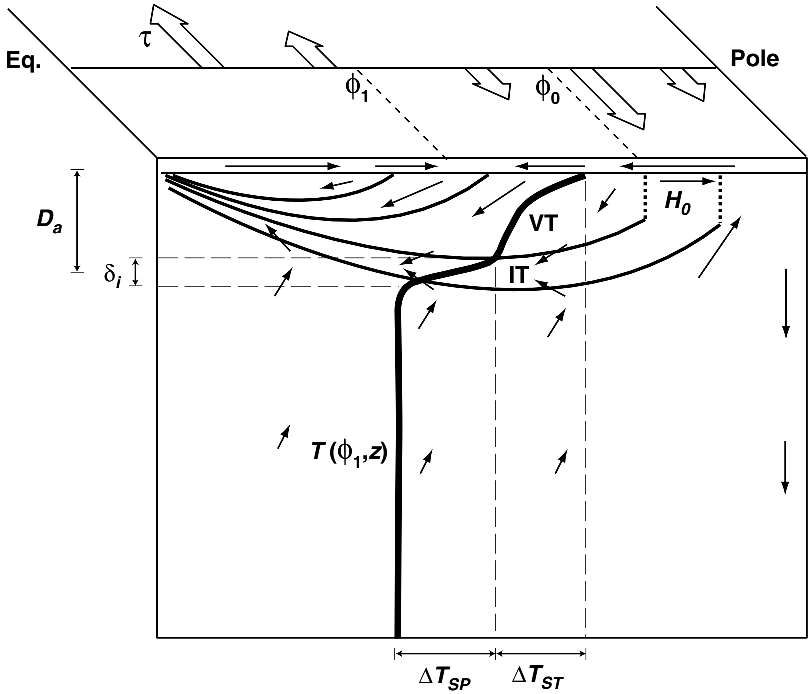
Temperature gradient across subpolar gyre was weaker than today.

Schematic of Main Thermocline



The region of the upper ocean where temperature (and density) change most rapidly. Loosely speaking, the thermocline is a front between warm subtropical waters and cold abyssal water.

The Thermocline



- Upwelling abyssal water sees an isotherm at the base of the ventilated thermocline.
- Little diffusion across the main thermocline.
- The thermohaline circulation is sensitive mainly to the meridional temperature gradient at *high* latitudes.

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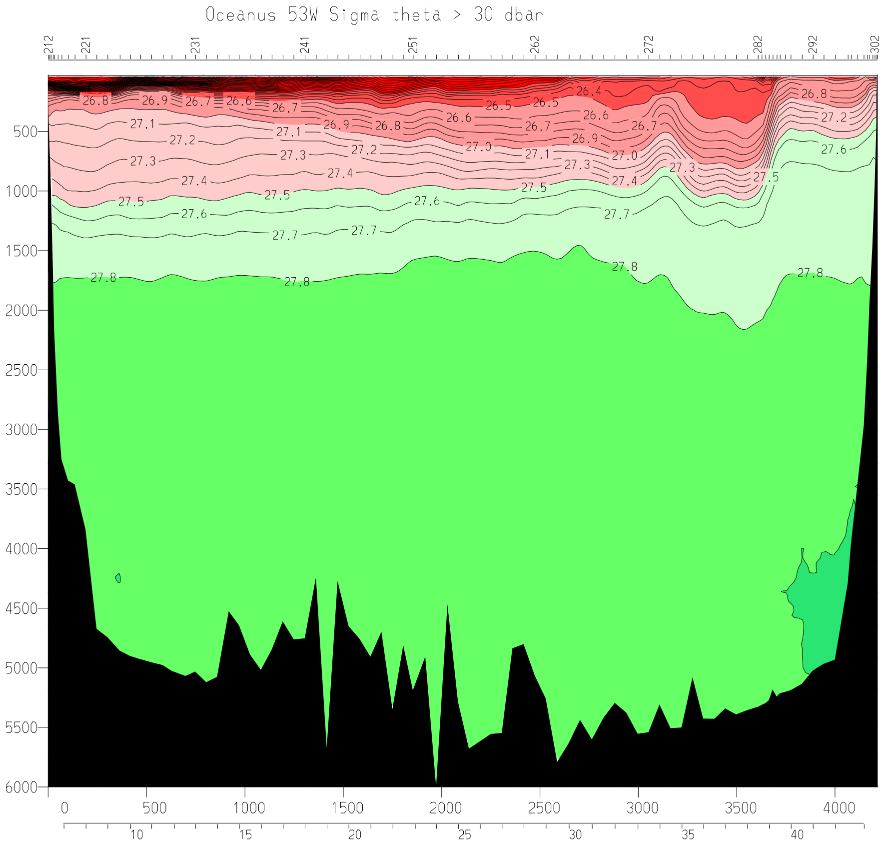
The Real Thermocline



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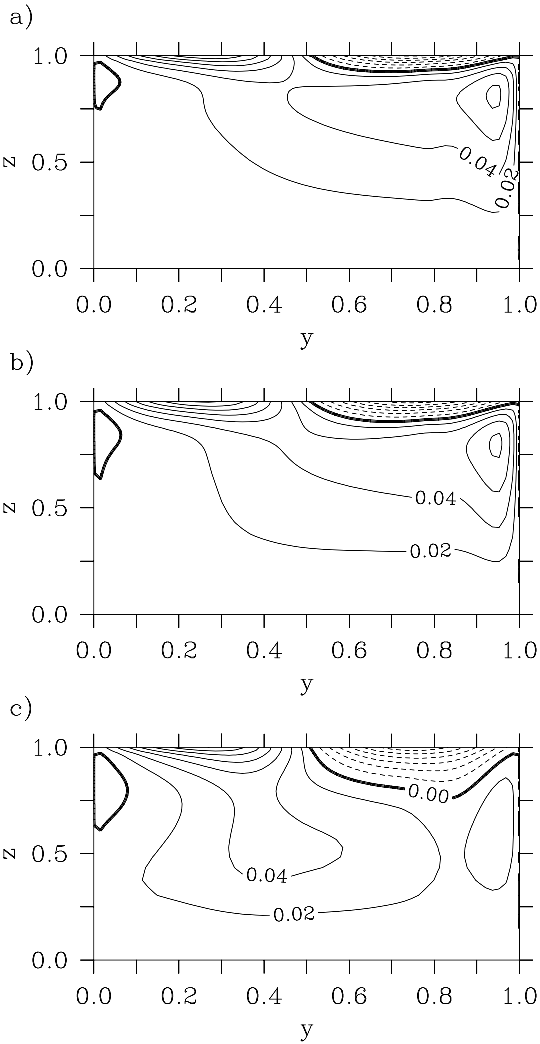
Overturning Circulation (in a model)



Control Case

No low-latitude meridional surface temperature gradient

No high-latitude meridional surface temperature gradient

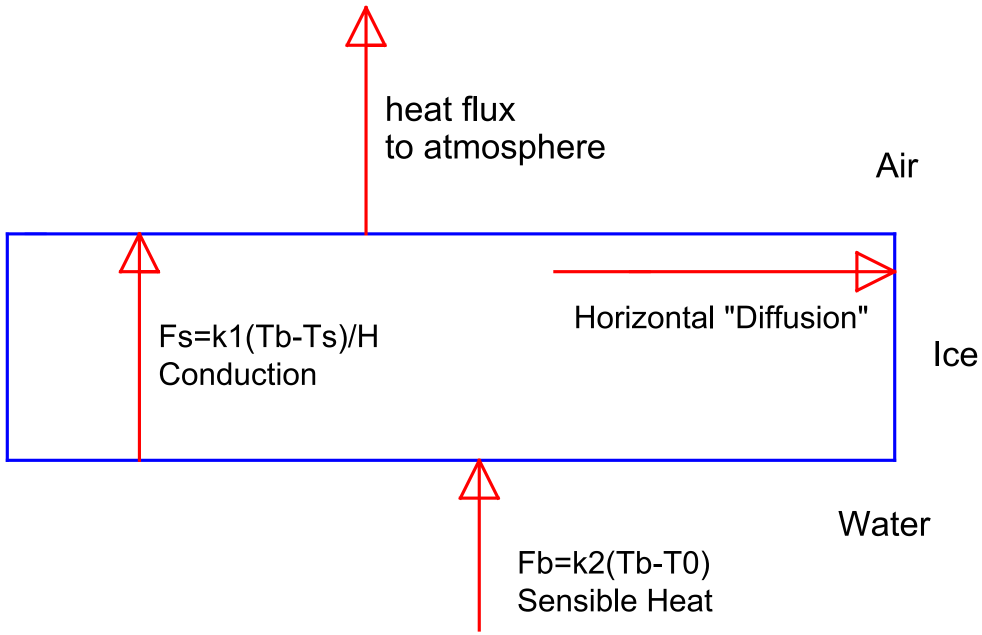
Model

- Couple a *three-dimensional* ocean model, with idealized geometry, to a simple ice model and an atmospheric Energy Balance Model (EBM).

  - The ice model is thermodynamic only. Grows or depletes ice according to temperature and thermodynamic balance at ocean surface. Acts to buffer the ocean from the atmosphere. When ice forms, heat loss from ocean to atmosphere is severely limited.
  - Atmospheric Model has a very simple radiation scheme, a simple scheme to transfer heat from the surface, and diffuses heat horizontally.

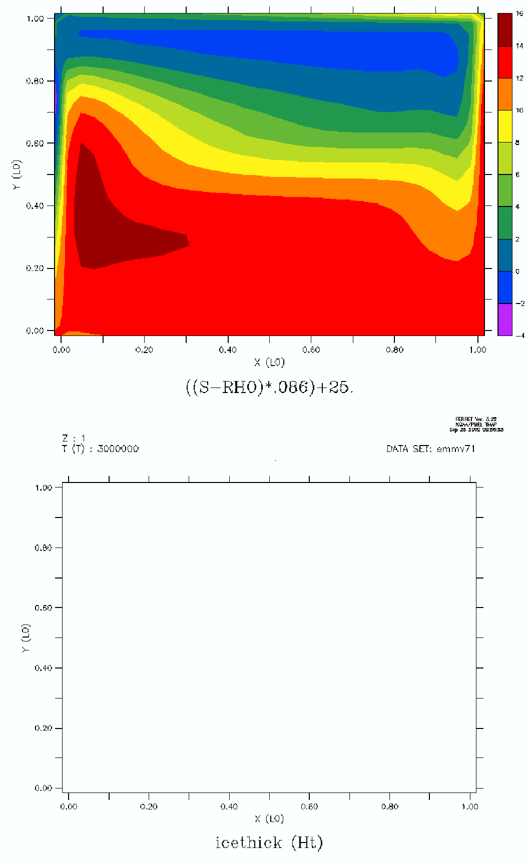
Create a ice-age world by changing the infra-red emissivity, to mimic lower CO_2 levels, and thereby lower overall temperature.

Schematic of Ice Model



Model predicts ice thickness. Ice *buffers* the ocean from a very cold atmosphere.

Typical Conditions in a 'Warm world'



Surface Conditions for a "Warm World"
SST in ºC (Top)
Sea Ice in Meters (Bottom)

Sea-surface temperature (SST)

Sea Ice

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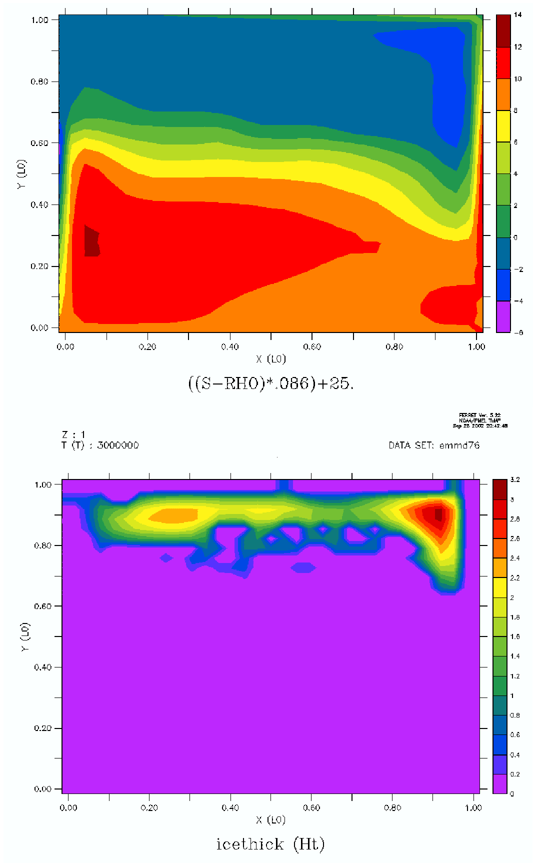
Typical Conditions in a Cold World

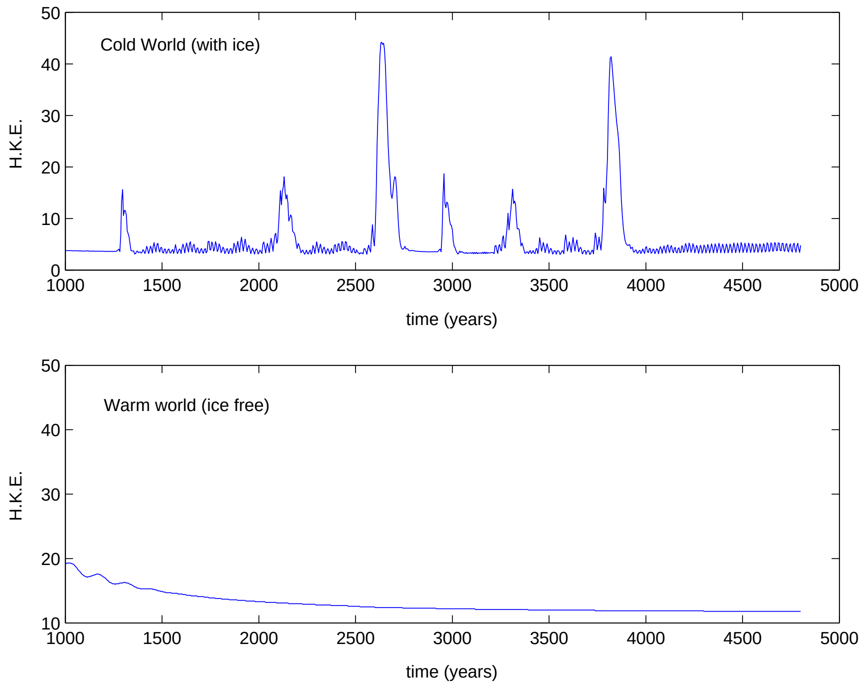


Surface Conditions for a “Cold World”
SST in ºC (Top)
Sea Ice in Meters (Bottom)

Sea-surface temperature (SST)

Sea Ice

Time Series in cold world and in a warm world



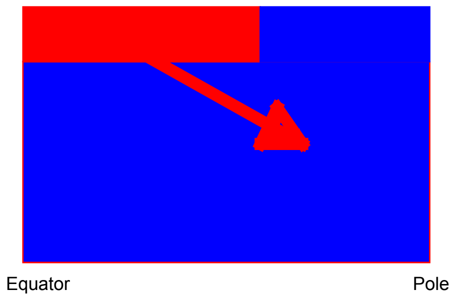


Fig. 9. Decoupled phase. The thermohaline circulation has halted and the deep water is slowly warmed “diagonally” by diffusion of heat. This phase takes from several hundred to several thousand years.

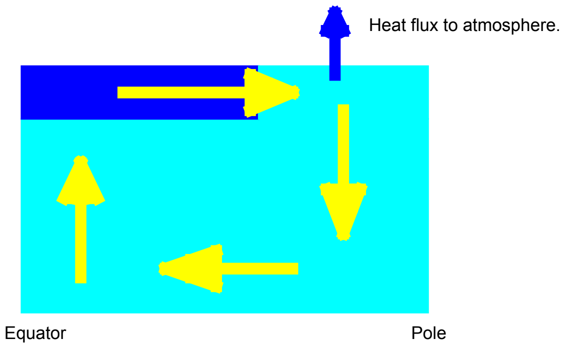


Fig. 10. Flushing stage. Convection causes the basin to rapidly overturn and the heat gained by diffusion (during the decoupled phase) is lost through the surface at northern latitudes. The heat loss through the surface rapidly warms the atmosphere over a period of 10 to 50 years. The rapid overturning lasts several hundred years.

Slow circulation leads to diffusive warming of ab yss. This leads to convective instability, high latitude warming and low latitude cooling.
(Winton 1995, others).

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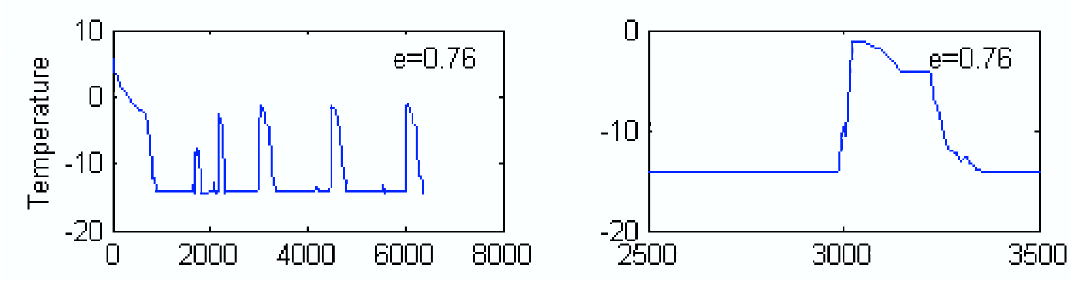
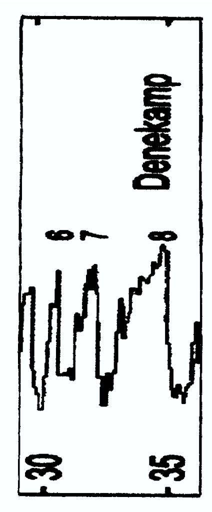
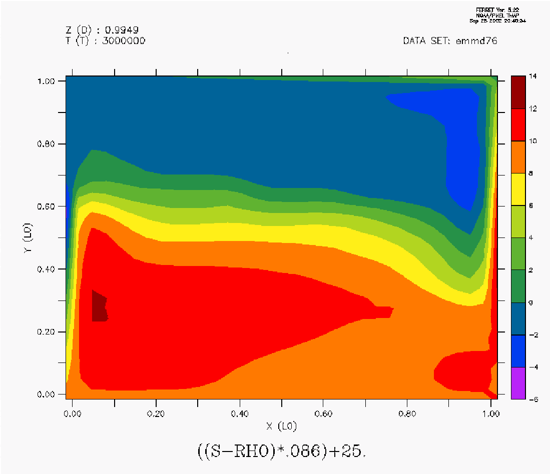
Characteristic Signature of a Warming event

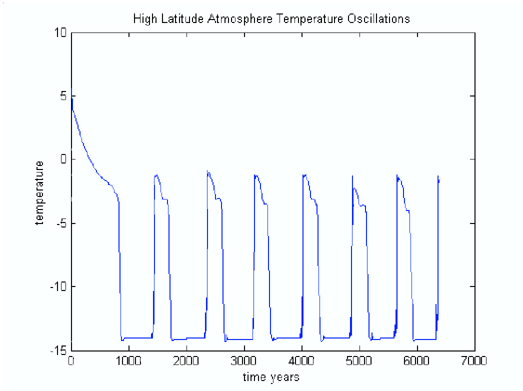




Observed δ^{18}O (left, from Dansgaard et al 1993) and modelled high latitude temperature (top). Events are characterized by a rapid warming and a slower return to glacial conditions. (Note that time axis is reversed in observed data graph.)

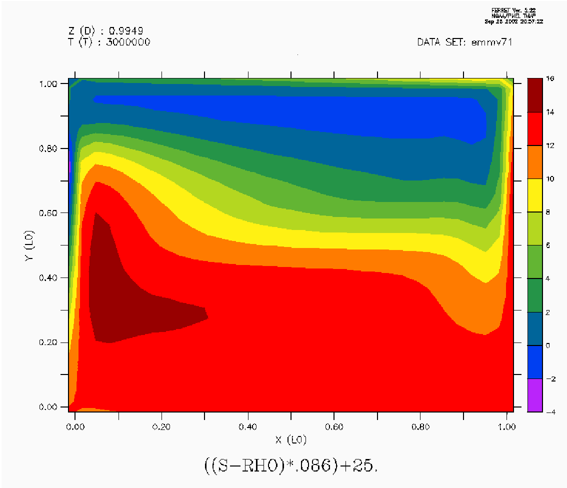


SST in a glacial time. Note almost uniform SST at high latitudes, with a strong gradient in high mid-latitudes.

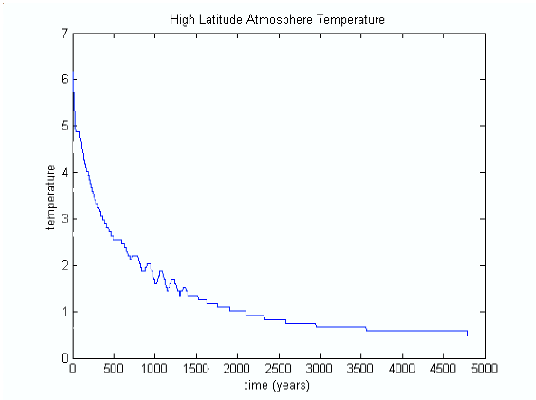


Time series of temperature.

Cold World.
Sea surface temperature (top) and high latitude atmosphere temperature time series (bottom).



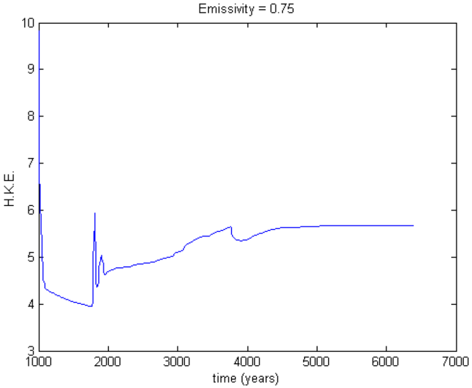
SST in a warm times. Relatively uniform meridional temperature gradient at all latitudes.



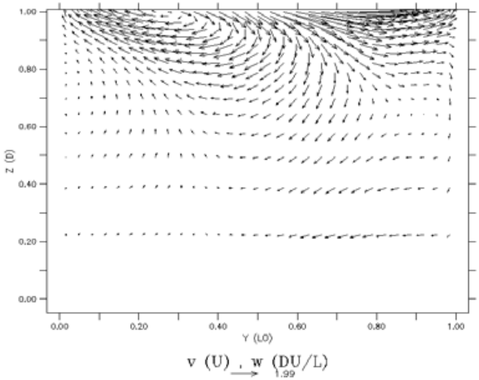
Evolves into a steady state.

Warm World.
Sea surface temperature (top) and high latitude atmosphere temperature time series (bottom).

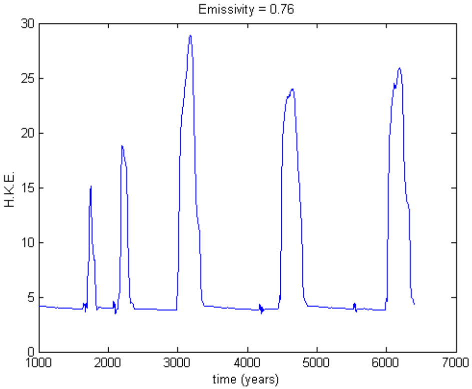
Warm World

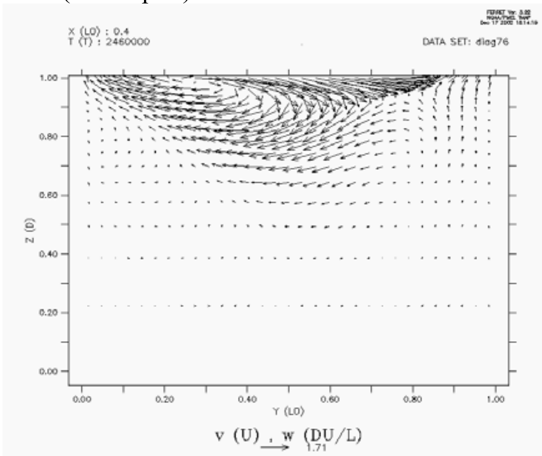


a. Basin integrated horizontal kinetic energy for e = 0.75

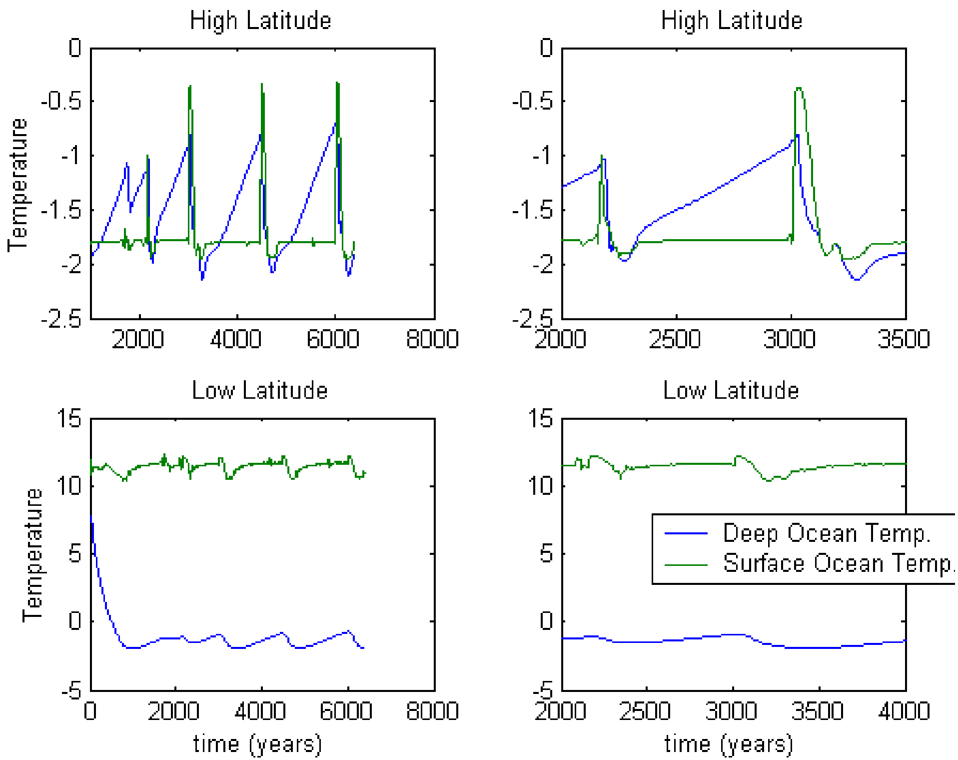


b. Meridional overturning circulation (vector plot) for e = 0.75

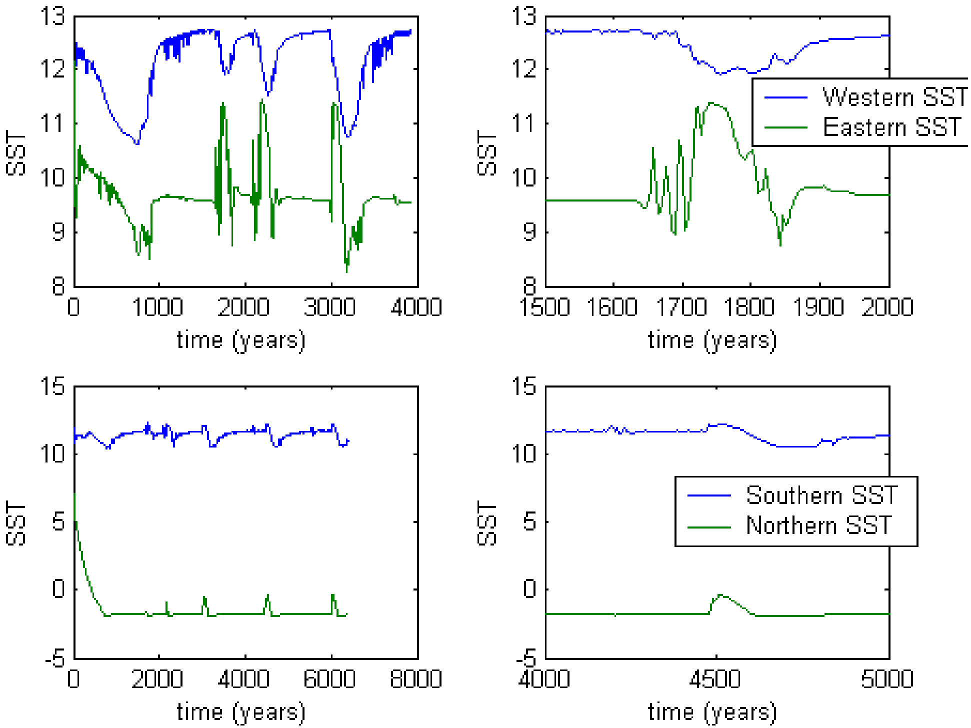




Cold World



Surface and abyssal SST variations at high and low latitudes. Abyssal temperature slowly warms between flushing events. Low latitude variations are smaller.



SST variations at two low latitudes, and at a north and south mid basin (bottom). Tropical western (eastern) SST cools (warms) during an event. SST changes are more abrupt at high latitudes.

Model and Observations

- In a cold climate, a weaker and more unstable thermohaline circulation, with brief warming periods.
  - Glacial climates typically more highly variable, with Dansgaard-Oeschger events on thousand year timescale.
- Saw-tooth shaped high-latitude temperature oscillations.
  - δ^{18}O has similar saw-toothed shapes for D-O oscillations (e.g., Dansgaard et al 1993), others.
- Weaker signal at low latitudes in model.
  - Weaker signal in southern hemisphere in observations (e.g., Blunier 1998), others.
- During a D-O cycle, southwestern cooling and southeastern warming.
  - Ruhlemann et al (1999) suggest western tropical SSTs fall and Eastern tropical SSTs rise as high latitude temperatures rise.

But

- This model has no southern hemisphere, and is too simple to see phasing variations among southern and northern hemisphere warming/cooling events.
- No evidence of abyssal temperatures rising between DO events (?)
- D-O events blur together in the historical record. The model events are quite distinct.
- Relation to Heinrich events is obscure.
- Role of nonlinear equation of state has not been explored. Role of salt has not been explored, and would be difficult to test.
- Detailed behavior is dependent on model vertical diffusivity.

What determines the eddy diffusivity of the ocean?

($64,000 question.)

$$w\frac{\partial T}{\partial z} \approx \kappa\frac{\partial^2 T}{\partial z^2}$$

Advection ≈ Diffusion

- Without a diffusivity the thermohaline circulation vanishes (nothing to work against).
  - The ocean *cannot* create its own eddy diffusivity without mechanical stirring. This is because the heating and the cooling of the ocean are at the same pressure. (Sandstrom 1908).
- Mechanical forcing can be by the wind, or the tides.
  - Without the moon, the climate might be quite different!!

Conclusions

- Various theoretical models suggest that a weak thermohaline circulation (THC) is more prone to variability than a strong one.

  If the THC is too weak, the abyss slowly warms by diffusion. Cold high latitude surface waters become unstable to convection, there is rapid onset of an overturning circulation, cooling the abyss and warming high latitudes. This circulation is not able to sustain itself, whence a return to ice cover.

- The glacial climate naturally has a *weak* thermohaline circulation.

  In part due to the natural cover of ice over the subpolar gyre: Reduces the meridional temperature gradient felt by the ocean in high latitudes, which in part sets the strength of the THC.

- Hence, the glacial THC and climate is prone to variability.

  Produce fairly rapid warming events, especially in high latitudes, resembling observed D-O oscillations. (Heinrich events?)

- More work needs to be done (!) to see if such a model can fit the (robust parts of the) paleo-record. *Need to simulate with full GCM.*

Predictability of Present-day Thermohaline Circulation

- Suppose we could observe the ocean very well. Could we predict the thermohaline circulation?
  - And would we care to do so?
- Thermohaline circulation may affect mid and high-latitude climate on interannual timescales — for example the North Atlantic Oscillation.
  - But, main determinant of ocean SSTs may be wind itself. So to predict the ocean, need to predict the atmosphere.

The Case that Climate is Predictable on Interannual-Decadal Timescales

- There is decadal variability in storm-tracks (the 'North Atlantic Oscillation').
- Deterministic slow timescales do exist in the system: the thermohaline circulation, the wind-driven circulation, sea-ice. Typically predictability is associated with and is on such a timescale.
- Given a history of SST in the Atlantic, some investigators are able to reproduce the trend in the NAO (Rodwell et al 1999) using a GCM.
- Predictability in some GCM studies (Griffies and Bryan 1997).
- El Niño seems to vary on decadal scales (e.g., 1930s were bereft of such events). This variation may be caused by slower variability in mid-latitude thermocline.

The Case that Climate is not Predictable on Interannaul-Decadal Timescales

- ENSO is a seasonal-interannual phenomenon, and not demonstrably predictable for longer than the next one, if that.
- Variations is the thermohaline circulation may have little effect on the SST, except at high latitudes.
- Although midlatitude SST anonalies may have a long life, the atmosphere may not only weakly influenced by them (Lau).
- Simple models of a stochastically forced atmosphere coupled to mixed layer ocean appear successful in interpreting various GCM experiments. These imply limited (seasonal at most) predictability. (Hasselman; Barsugli, Bretherton & Battisti).

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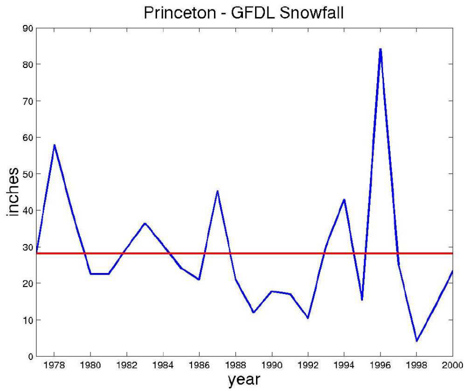
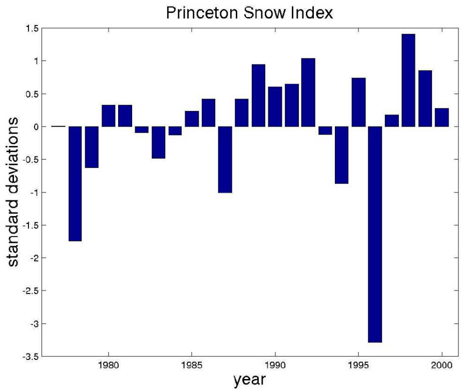
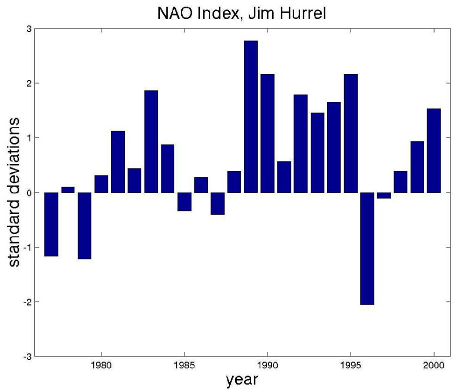
The Princeton Index (!)





Princeton Snowfall (above), the Princeton Snowfall index (top right), and the NAO index (bottom right).



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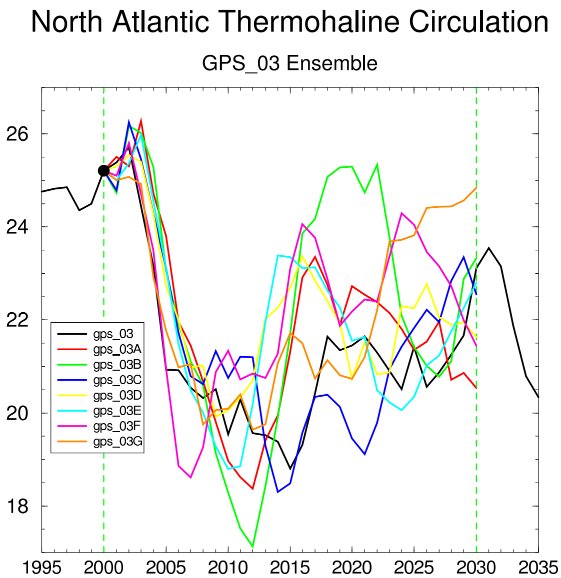
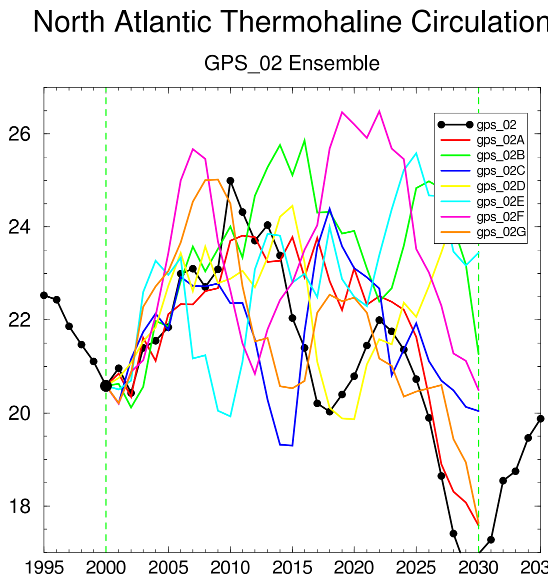
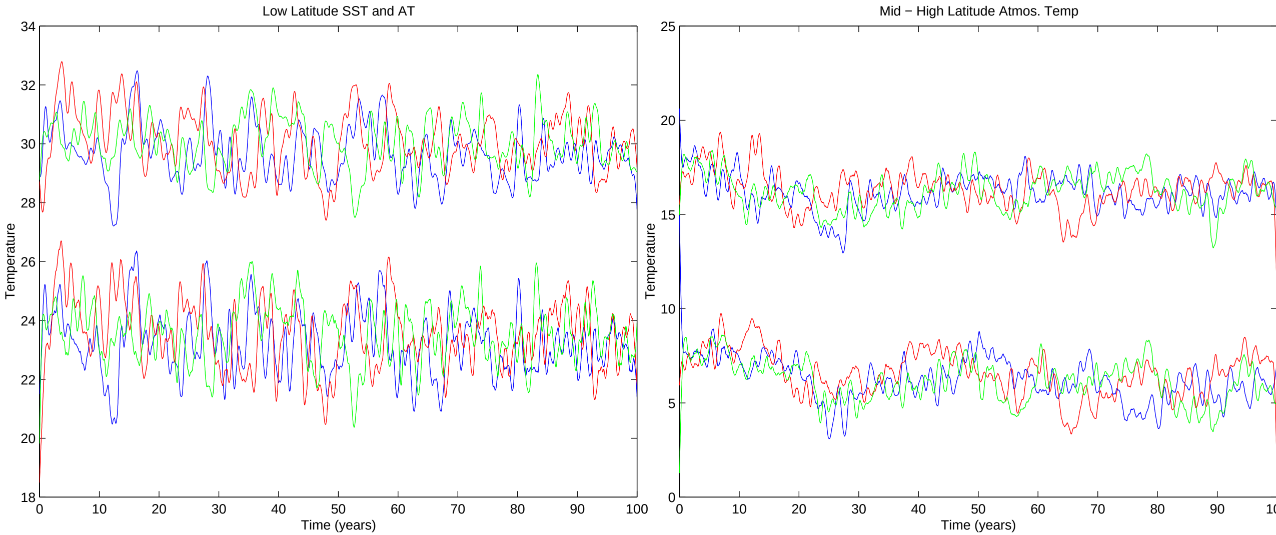
Thermohaline Predictability (T. Delworth, S. Griffies)





Low and high latitude SST predictability



The Moral of the Story

- Atmosphere-ocean-climate dynamics has a nice confluence of pure and applied problems, esoteric and practical.
  - Justify equally to a Program Manager or a Platonic philosopher.
- Its a complicated system. Big models (GCMs) are vital, but they don't make theory less important. Rather, theory becomes *even more important*, both to understand the system and to improve the models.